GLOBAL
; up · **.** down · **,** **/** left/right · **ENTER** select · **`/DEL** back · **{ }** page · **b** screenshot · **h** help (Help links: **g** glossary+math · **m** user guide · **s** sat history · **t** tech help · **l** learn theory)

HOME
ENTER opens item; menu scrolls: Satellites, Next Passes, Passes, Track, World Map, Sun/Moon, Space Wx, Weather, Activations, Overhead now, QRZ Lookup, Location, Update, Settings, Log, Messages, About, Charge/Sleep

SATELLITES
f favorite · **v** favs-only · **n** new GP sat · **x** del manual sat · **e** EQX table · **k** OSCARLOCATOR · **3** 3D globe · **2** sat-to-sat · **o** orbital · **s** sim · **t** transponders · **d** 10-day · **i** illum · **ENTER** passes · right edge: dot = AMSAT heard, square = telemetry, ring = not heard

SAT-TO-SAT (Sats → 2)
Windows when selected sat + a 2nd fav are BOTH above your horizon (start, dur, peak el each). **n** next sat · **r** recompute · ` back

LIVE GPS POS (Location → v)
DMS lat/lon (max precision) + decimal, alt, grid, speed, course. Rovers/portable. ` back

3D GLOBE (Sats → 3)
Wireframe Earth, auto-follows selected sat. Graticule, coastline, day/night terminator, QTH, favs, selected sat + footprint + ground-track trail. **g** DX 2nd location (grid) + footprint · **G** clear · **arrows** turn · **ENTER** re-follow · ` back

EQX TABLE (Sats → e)
Equator-crossing UTC + longitude (W-positive) for OSCARLOCATOR, next 3 days. **;/.** scroll · **d** asc/desc node · **r** recompute · ` back

OSCARLOCATOR (Sats → k)
Live azimuthal-equidistant plot: sub-point, footprint, QTH range ring + full ground track (AOS/LOS). **m** toggle polar (default, auto N/S, flips at equator) / QTH-centred · ` back

DX DOPPLER TABLE (Mutual → d)
RX/TX dial freqs for BOTH stations every 30s across a mutual window. Two lines/step: me (green) + DX (cyan). **t** cycle transponder · **m** mode: true rule / fixed DL / fixed UL · **a** anchor dial (me/DX RX/TX) · **,//** in fixed mode step anchored dial to round 1 kHz (else passband 1 kHz) · header shows pb +/- from center · **;/.** scroll · ` back

NEXT PASSES (favs)
ENTER track · **m** world map · **t** sky-at-a-glance timeline (bars by elev) · **r** refresh · **z** deep-sleep until AOS

PASSES (sel)
***** = optically visible. **;/.** select · **d** detail · **t/ENTER** track · **n** add TX · **r** recompute · **x** mutual-DX · **v** 10-day chart · **V** visible-pass list · **i** illum · **g** grids · **w** US states · **e** DXCC

TRACK (sel)
` exits to previous screen & KEEPS radio/rotator tracking (green RAD/ROT/R+R in header); **r/o** to stop. **m** TUNE/CAL · **d** tune mode (FULL/DL/UL/hold) · **t** next TX · **n** jump to beacon · **c** CTCSS · **N** sat note · **k** CW both legs (linear) · **r** radio · **o** rotator · **p** polar · **a** point-here arrow · **z** big readout · **y** tilt on/off (ADV) · **f** Manual · **l** log QSO · **v** voice memo (SD) · **g** grids · **w** states · **e** DXCC now · **ENTER** save cal

BIG READOUT (z from Track)
Big RX/TX + az/el + tune mode (follows Track). Radio+rotator keep tracking · **,//** tune · **s/x** step/ctr · **m/d** mode · **t** TX · **n** beacon · **k** CW (linear) · **r** radio · **o** rot · **y** tilt · **l** log · **z/`** back

MANUAL (no radio)
Fix one leg, read Doppler freq to tune the other by hand. **u** toggle HOLD/TUNE leg · **,//** passband (linear) · **m** CAL · **t** next TX · **z** big view · **l** log · **p** polar · **g** grids · **w** states · **e** DXCC · `**/f** Track

MANUAL BIG (z from Manual)
HOLD/TUNE legs in big digits · **u** swap leg · **,//** tune · **s/x** · **m** CAL · **t** TX · **z/`** back

TRACK · TUNE
,// tune -/+ · **s** step 100/1k/5k · **x** recenter · tilt to tune if enabled (ADV)

TRACK · CAL
,// downlink · **;/.** uplink · **s** step 10/100/1k · **x** zero

WORKABLE GRIDS / STATES / DXCC
Footprint coverage (per-pass union or live now), count on a cyan line: **g** 4-char grids · **w** US states+DC · **e** DXCC (all 340, hybrid polygons+points). On grids, **f** prefix filter (EM/EM2/EM21, upper-cased), **c** clear. **;/.** & **{/}** scroll · ` back

POLAR / PASS DETAIL
Pass detail: **p** polar of this pass. Polar: **l** log QSO · **v** voice memo (SD) · **p/ENTER/`** back

SUN / MOON
Graphic sky-dome (Sun/Moon glyphs by az/el) · **g** graphic/list · **;/.** pick Sun/Moon · **o** rotor track on/off · **s** sky sources · **x** stop · ` back

SKY SOURCES (Sun/Moon → s)
Planets (cyan dots) + strong radio sources (orange +): Cas A, Cyg A, galactic centre, Crab, Virgo A, on a sky dome. Antenna-pointing / RF reference. **;/.** select · ` back

SPACE WX (menu)
Solar 10.7cm flux + planetary Kp + A index + aurora likelihood (from Kp), labelled & colour-coded, with HF/sat operating outlook & data age · **r** refresh (WiFi) · ` back

WEATHER (menu)
Current conditions + multi-day forecast for your site (Open-Meteo). Refreshes on entry (WiFi) & with Update. Units in Settings · **r** refresh · cached offline · ` back

QRZ LOOKUP (menu)
Callsign lookup via QRZ.com XML (needs QRZ XML subscription + user/pass in Settings → Network). **ENTER** type call → name/addr/grid/class. WiFi req'd · ` back

ACTIVATIONS (menu)
Upcoming sat activations scheduled on hams.at (roves, grid/special ops). List: date, call, sat, grid (* = your own entry). **;/.** move · **ENTER** detail (UTC times, mode, freq, comment) · **n** add your own sked (offline OK), **e** edit a * entry · in detail **a** sets a SKED reminder (T-60/30/10 beeps + flash at start, separate from AOS alarm); **c** on list clears it · **r** refresh · ` back. Cached to card — last list shows offline; WiFi to refresh

OVERHEAD NOW (menu)
Snapshot of every catalog sat above the horizon right now, sorted by elevation, with az + rise compass dir (high=green, low=yellow) + count up/scanned. **r** rescan (this instant) · **;/.** scroll · ` back

TRANSPONDER DB (Sats → t)
Sat's transponder/beacon entries, two lines each: **D**=downlink+mode line, **U**=uplink+tone/inv line. **;/.** select (* = manual) · **x** del manual (2x) · ` back

ORBITAL ANALYSIS
,/ 9 pages: Info / Live / Next pass / Ground track / Doppler / Nodal / Sun-Beta / Pass outlook / Orbit position · **r** recompute · Doppler **f** sets beacon freq

SIMULATION
,/ step time · **;/.** step size · **m** world-map view (sub-point + footprint) · **x** reset to now · ` back

LOCATION
e/**o**/**a** lat/lon/alt · **g** grid · **p** GPS on/off · **s** GPS source · **c** set clock · **v** live position (DMS) · **ENTER** GPS sky plot

GPS SKY PLOT
Live GNSS by az/el, coloured by signal (green=strong, grey=weak) · ` back

WORLD MAP
All footprints + night-side shading · **f** highlight favorite · **c** recenter on QTH/0° · yellow=sunlit cyan=eclipse · ` back

SCHEDULES
10-day: **;/.** scroll +/-1 day (fills full days), **r** recompute · Illum: **,/** scroll +/-60 days · Mutual: **;/.** scroll

LOG
Menu (scrolls): **ENTER** new QSO / browse / export ADIF / LoTW upload / Cloudlog upload / voice memos / notes / awards · **Awards**: QSO/grid/state/DXCC totals (states+DXCC derived from grid), **g**/**s**/**d** scrollable worked lists, **ENTER** per-sat · List: **;/.** scroll, **ENTER** edit · Entry: **;/.** field, **ENTER** edit, **s** save, **x** x2 delete · editing a QSO re-arms its upload; extra **LoTW**/**Cloudlog** rows (ENTER toggles) override that

SIGN & UPLOAD TO LoTW (Log → Sign & upload)
Uploads sat QSOs direct to ARRL LoTW over WiFi. Needs SD + your LoTW key (lotw_key.pem/lotw_cert.pem in /CardSat/ — make them from your TQSL .p12 with tools/lotw_cert_converter.html in a browser; in Settings pick DXCC → primary (state/province/...) → secondary (county/city) + zones + IOTA, all gated pickers w/ type-to-filter) — see manual §8. **u** upload · **a** re-send all · ` back

UPLOAD TO CLOUDLOG (Log → Upload to Cloudlog)
Uploads sat QSOs to a self-hosted Cloudlog/Wavelog over WiFi (an alternative to LoTW — Cloudlog can forward to LoTW). Set **URL** + read-write **API key** + **station ID** in Settings. **u** upload · **a** re-send all · ` back

VOICE MEMOS (Log → Voice Memos)
Browse newest-first (date/time/sat/len). **ENTER** play · **n** new · **d** del · **r** refresh · record via **v** on Track. SD req'd

NOTES (Log → Notes)
Free-form text notes (.txt in /CardSat/notes/, newest-first w/ date/time). Browser: **ENTER** open · **n** new · **d**+ENTER del. Editor (commands use **Fn** so **;.,/** type): type freely, ENTER=newline, DEL=backspace · **Fn+,/Fn+/** cursor L/R · **Fn+;/Fn+.** up/down · **Fn+s** save · ` exit (auto-saves)

LORA MESSAGES (Home → Messages)
CardSat-to-CardSat broadcast chat (Cap LoRa). Same freq/SF/BW = same group. Set **region** (US 33cm / EU 70cm / JP 430) in Settings for a legal default freq. **n** write/send · **;/.** scroll · **r** retry · ` back. **Rx always on**: new msgs show an envelope+count badge in the header on any screen, plus a banner (opt-in beep) — set **Msg notify** (off/banner/banner+beep) in Settings. Needs RadioLib build. Untested HW

UPDATE
k/ENTER GP+clock+AMSAT+space-wx+weather · **f** fast (GP+AMSAT+favs' TX) · **a** cache all TX (auto-reboots) · **w** WiFi only

SETTINGS
,/ change · **ENTER** edit/toggle · **s** scan WiFi · opt. 2nd WiFi (field fallback) · reset = ERASE

GP SOURCE
AMSAT / **CelesTrak** JSON-PP category / **Custom URL** · **;/.** move · **ENTER** select

ROTATOR (manual)
,/ az · **;/.** el · **s** step · **x** stop · ` back

NETWORK SERVERS
rigctld PC drives rig (VFOA=DL/B=UL) · **rotctld** PC drives GS-232 · **rigctl** drives remote rig · **Web** opt-in mobile page (no auth, trusted LAN): live sky plot, Doppler readout, tap-to-copy freqs, visible-pass list + AOS alerts, radio/rotator control

EDIT
type · **DEL** backspace · **ENTER** ok · ` cancel

ABOUT
Build/version, IP, free heap and diagnostics (read-only). **l** License & credits (disclaimers, data sources, support AMSAT) · **z** Zap the Sats (mini game: **,/** move, SPACE fire, ENTER start).